

User's Guide for Series 60 SIP Plug-in

Version 1.0; May 17, 2004

SIP

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Change History

May 2004	Version 1.0	First issue of the document
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1 About this document

Series 60 SIP Plug-in consists of SIP Server Emulator, SIP Extension for Series 60 Terminal Emulator, an example SIP application, and the relevant documentation.

This User's Guide provides:

- The settings necessary to configure a SIP development environment.
- Information on administrative issues such as starting and stopping the SIP Server Emulator, configuration and, log reading.
- Information on using the SIP Server emulator's graphical user interface (GUI).

1.1 Typographical conventions

The following typographical conventions are used throughout this guide:

Bold	Bold is used to indicate windows and their elements, menu items, and button names.
<i>Italic</i>	Italics are used when referring to a manual.
<code>Courier</code>	Courier is used to indicate parameters, file names, processes, commands, directories, and source code text.
	Pipelines are used to separate menu items within a path.

Note: Notes to the reader are presented in this format.
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1.2 Audience

This guide is for developers who create SIP applications and use Series 60 SIP Plug-in. Before getting started, users of this guide should be familiar with the following:

- Symbian 7.0 application development

1.3 List of Tables/Figures

Figure 1: Series 60 SIP Development environment

Figure 2: Ethernet Plug-in network configuration

Figure 3: Startup window of the SIP Server Emulator

2 Overview of Series 60 SIP Plug-in

Session Initiation Protocol (SIP) is a protocol that initiates sessions between terminals. With Series 60 SIP Plug-in, initiation is established through the SIP Server Emulator. When a session is initiated, terminals continue interacting without the SIP Server Emulator.

The SIP Server Emulator combines the functionality of SIP Proxy Server and SIP Registrar Server. It is the forwarding center that delivers SIP messages to its recipients. Incoming and outgoing SIP messages can be logged into a log file. SIP Server Emulator can also be used in automatic response mode. When the SIP Server Emulator receives a SIP message to be forwarded, the SIP Server Emulator responds with a specified SIP message and does not forward the incoming SIP message. This automatic response mode is used mainly for testing SIP error messages.

SIP Extension for Series 60 SDK allows SIP features in Terminal Emulator. SIP Extension makes it possible to send and receive SIP messages to and from the Terminal Emulator.

When combined, Series 60 SIP Plug-in and Terminal Emulator form an environment in which you can test your SIP applications. The example SIP application Warship is used for testing Series 60 SIP Plug-in.

3 Before you start

3.1 Platform requirements

Series 60 SIP Plug-in has been tested in the following operating systems: Microsoft Windows 2000 and Windows XP.

Series 60 SIP Plug-in works under Java™ Platform Standard Edition version 1.4.1_02 or higher. The required Java™ version is installed during Series 60 SIP Plug-in installation.

3.2 Environment setup

This section describes how to set up Series 60 SIP Plug-in (SIP Development environment for Series 60 SDK 2.0). Building the whole environment consists of a series of actions that happen in a specific order. Those steps include installing all the required software components and configuring them so that they work together.

As a result of the actions described in this section, the developer can access the whole environment in one PC (see figure 1).

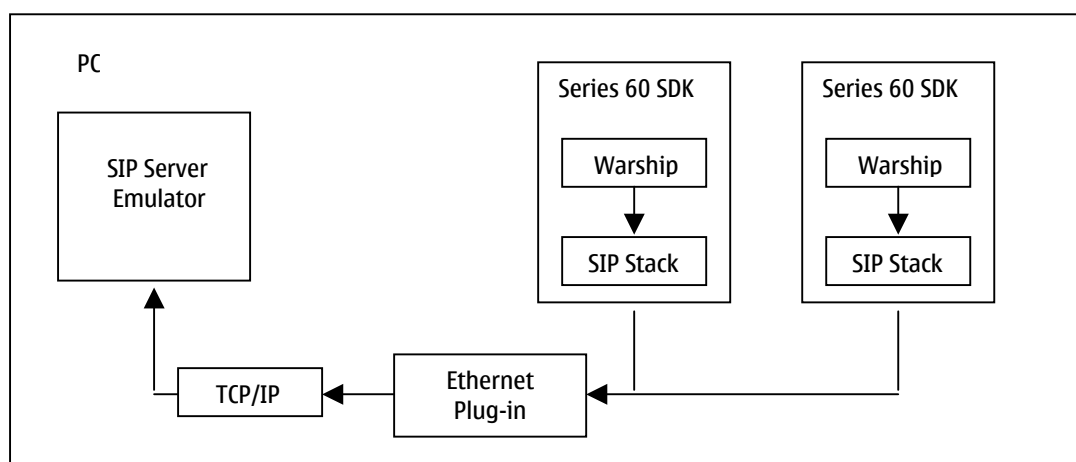


Figure 1: Series 60 SIP Development environment.

The PC will have:

- A fully functional, correctly configured, "ready to run" SIP Server (that is, SIP Server Emulator) with SIP message logging and automatic response -features.
- Two Series 60 SDK 2.0 environments with
 - the SIP extension (the SIP API) that enables SIP features in Terminal Emulator applications
 - an example SIP application, the Warship game, which makes it possible to check that the environment is correctly configured
- Ethernet Plug-in for Series 60 SDK 2.0. This enables network communication between the Terminal Emulators and the server.

The 7 step setup procedure is introduced in this section. Each step is described in more detail in the following chapters. The setup procedure propagates as follows:

- Install Series 60 SDK 2.0
- Install the SIP Plug-in
- Build the example SIP application
- Install Ethernet Plug-in for Series 60 SDK 2.0
- Make a copy of Series 60 SDK 2.0
- Configure network settings for Series 60 SDK 2.0 B
- Test the environment

3.2.1 Install Series 60 SDK 2.0

If not installed already, install Series 60 SDK 2.0. You might want to use MS Windows subst command to assign a path with a drive letter. For example:

```
subst S: C:\Symbian\Series60_v20
```

Note: Do not use subst in Windows XP, the included version of Ethernet Plug-in cannot be installed on the substed drive.

3.2.2 Install the SIP Plug-in

Install SIP Plug-in by executing `setup.exe` from **Series 60 SIP Plug-in** package. During installation, choose **Series 60 SIP Plug-in 1.0** install set. The Series 60 SIP Plug-in installer will also install JRE 1.4.1_02 under SIP Plug-in installation directory. JRE installed during SIP Server Emulator installation will not be available for applications other than SIP Server Emulator itself.

Note: Restarting the PC is required to enable the SIP Server Emulators SIP message logging feature.

3.2.3 Build the example SIP application

It is recommended to clean the current environment before update:

- Run `[SDK_PATH]\Columbus\clean.cmd`
- Remove COLUMBUS directory from `[SDK_PATH]\Epoc32\BUILD\`
- Build the application again, see section 3.2.3.1 or 3.2.3.2

3.2.3.1 Build the example SIP application with Visual C++

In Command Prompt, go to the [SDK_PATH]\Co lumbus directory and execute

```
build wins udeb
```

[SDK_PATH] is the SDK installation path. If subst used [SDK_PATH] is the assigned virtual drive. Look for any error messages from the build output.

Note: The build command requires the environment for Microsoft Visual C++ tools. If environment variables are not previously set, they can be set by running vcvars32.bat from Microsoft Visual Studio installation directory\VC98\Bin. The build command must be then executed in the same command prompt as the vcvars32.bat so that the environment is used.

3.2.3.2 Build the example SIP application with CodeWarrior

In Command Prompt, go to [SDK_PATH]\Co lumbus directory and execute

```
build winscw udeb
```

[SDK_PATH] is the SDK installation path. If subst used [SDK_PATH] is the assigned virtual drive. Look for any error messages from the build output.

3.2.4 Install Ethernet Plug-in for Series 60 SDK 2.0

If not already installed, install Ethernet Plug-in for Series 60 SDK 2.0. The Ethernet Plug-in is delivered with the Series 60 SIP Plug-in installation package, but it is not installed during SIP Plug-in installation. Check the Forum Nokia web pages for the latest version. Install the Ethernet Plug-in and follow the installation program instructions. Configure network settings as following:

- Select Network adapter in promiscuous mode.
- IP address = a free IP for your network. To do this, contact your network administration.
- Subnet mask = 255.255.255.0.
- Default Gateway = IP address of the PC if no external network present or if IP forwarding enabled for used Ethernet interface. In other case, address of gateway of that network for which PC is connected.

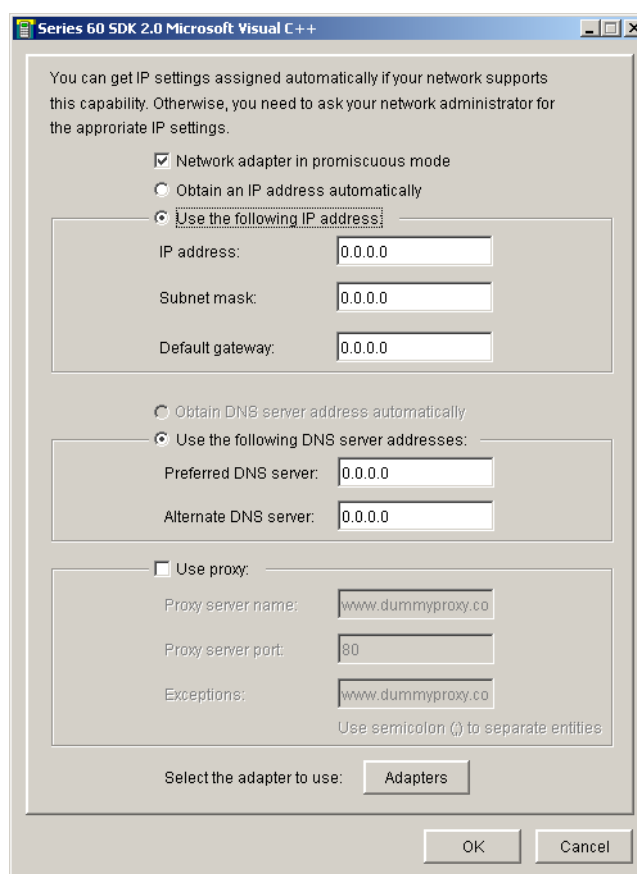


Figure 2: Ethernet Plug-in network configuration

Note: If you have Series 90 SDK installed, it may prevent Ethernet Plug-in installation.

3.2.5 Make a copy of Series 60 SDK 2.0

Make a copy of the whole [SDK_PATH] –directory and place it somewhere on the PC's hard drive. The [SDK_PATH]-directory can also be renamed and copied in the same location as the original [SDK_PATH]-directory.

You might want to use the MS Windows subst command to associate the new path with a new drive letter. See section 3.2.1.

Please note that from this point on in this document, the original Series 60 SDK 2.0 installation is called “Series 60 SDK 2.0 A” and the copy is called “Series 60 SDK 2.0 B”.

3.2.6 Configure the network for Series 60 SDK 2.0 B

Network configuration for Series 60 SDK 2.0 A is done when configuring Ethernet Plug-in network settings. Series 60 SDK 2.0 B must be configured manually.

Run [SDK_PATH_B]\Epoc32\tools\network_config.bat and modify following field:

- IP address = a free IP for your network. To do this, contact your network administration. Use a different IP address than in Series 60 SDK 2.0 A.

3.2.7 Test the environment

To test the Series 60 SIP development environment, run the example SIP application, Warship. The SIP features of the Series 60 Terminal Emulators must be configured before running the Warship.

Start SIP Server Emulator (GUI) from Start menu. Before starting the server, you must configure the domains to be used. Find out the PC's IP address and add it to the domains list. Removing the default domains (domain1 and domain2) is not required. Start SIP Server Emulator by pressing the Start server button on the toolbar. SIP Server Emulator should now accept SIP requests.

In Visual C++ version:

Run [SDK_PATH]\Epoc32\release\wins\udeb\epoc.exe

Run [SDK_PATH_B]\Epoc32\release\wins\udeb\epoc.exe

In CodeWarrior version:

Run [SDK_PATH]\Epoc32\release\wincsw\udeb\epoc.exe

Run [SDK_PATH_B]\Epoc32\release\wincsw\udeb\epoc.exe

On both Series 60 Terminals Emulators, find "SIP" icon and set the SIP configuration. The correct configuration is as follows:

Provider:Provider

Service profile:IETF

Default access point:LAN_IAP_1

Public user name:player1@realm

Use compression:No

Registration:When needed

Use security:No

Proxy server:

Proxy serv. address:10.0.0.10

Realm:realm

User name:player1

Password:player

Allow loose routing:Yes

Transport type:UDP

Port:5060

Registrar server:

Registrar serv. addr.:10.0.0.10

Realm:realm
User name:player1
Password:player
Transport type:UDP
Port:5060

10.0.0.10 should be replaced with the PC's IP address. On the other Series 60 Terminal Emulator, the name "player1" should be replaced with another name, such as "player2". "realm" can be replaced with any text. Player1 and player2 can have different realms.

On both Series 60 Terminal Emulators, add an entry on the Phonebook. The entry will contain the opponent for the Warship game. The entry must contain at least the first name and an URL field. The URL field must contain the SIP address, such as sip:player2@realm. On the other emulator, the phonebook entry is like sip:player1@realm. Exit Phonebook.

On both Series 60 Terminal Emulators, find the Warship icon and start the application. Configure the settings in the following way:

Nickname:player1
SIP Profile:Provider

On the other Series 60 Terminal Emulator, the name "player1" should be replaced with the name used in the SIP configuration. You should now be able to start the game session successfully. It should also be possible to see incoming and outgoing SIP messages on the SIP Server Emulator's GUI.

4 Using the SIP Server Emulator

Start the server UI by running **Start | Programs | Nokia Developer Tools | Series 60 SIP Plug-in 1.0 | SIP Server Emulator** on your Start menu.

The following window appears.

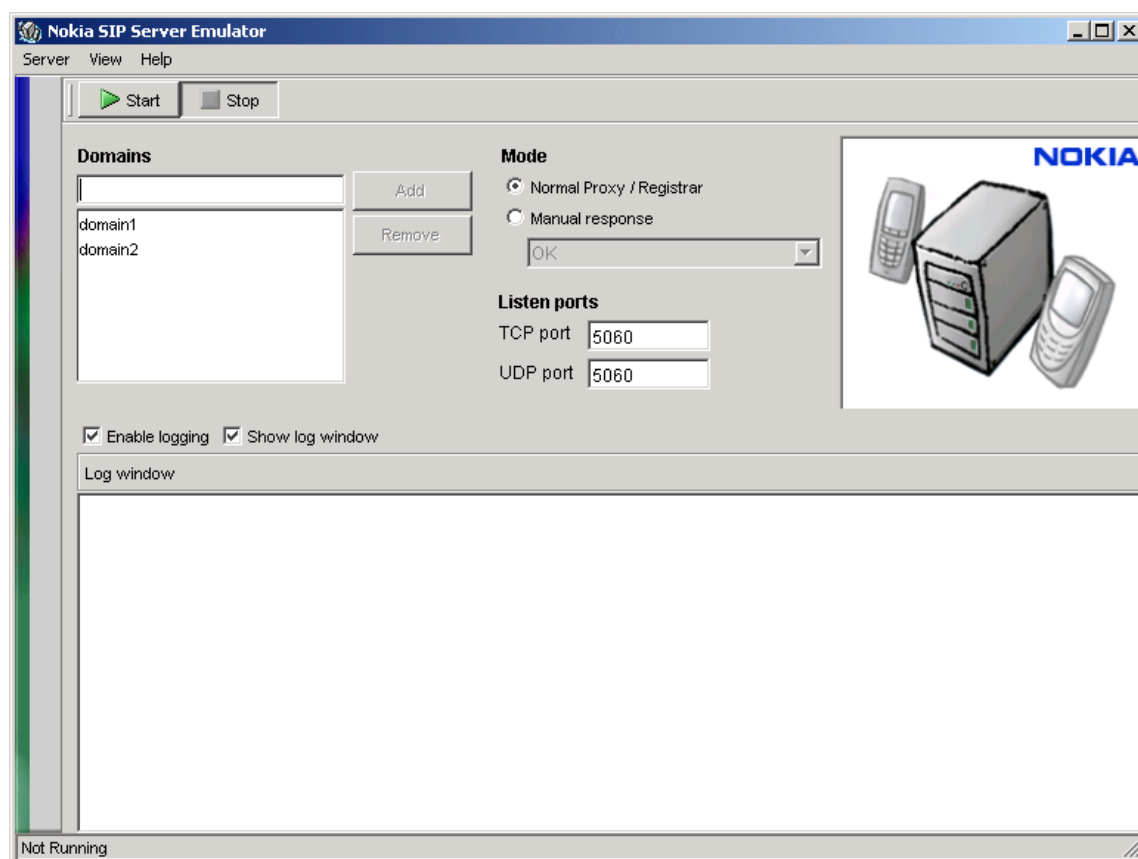


Figure 3: Startup window for SIP Server Emulator

4.1 Configuring the SIP Server Emulator

SIP Server Emulator must be configured before starting up. The configuration is set in the main window.

4.1.1 Domains

Before starting up SIP Server Emulator, a domain name must be defined. The domain name must match the Realm name used in Terminal Emulator. Multiple domain names can be defined. A single domain name cannot be added twice. Domain name cannot contain empty spaces. Empty spaces are removed automatically before adding a new domain name to the list. New domain names can be added while the SIP Server Emulator is running. Domain configuration becomes operative by pressing **Add** button. The SIP Server Emulator must be stopped before removing domain names from the list. To remove a domain name, select the domain name to be removed and press the **Remove** button.

4.1.2 Listen ports

Additionally, TCP and UDP ports must be defined before starting up SIP Server Emulator. Ports must be numerical and cannot contain empty spaces. Empty spaces are removed automatically before starting up SIP Server Emulator. Default value for TCP and UDP port is 5060. Port configuration can be modified only when the SIP Server Emulator is not running. Port configuration becomes operative when the SIP Server Emulator is started.

4.1.3 Mode

SIP Server Emulator has two modes. The mode can be changed while the SIP Server Emulator is running. The Normal Proxy / Registrar mode is normal SIP Registrar and SIP Proxy Server, which registers SIP addresses and forwards SIP messages. By selecting Manual response mode, the SIP Server Emulator starts to respond with selected responses instead of forwarding SIP messages. Manual responses can be chosen from the response combobox that contains a few SIP response scenarios. A SIP response scenario contains message chains with one or more SIP messages, for example, 100 Trying, 404 Not Found.

Some scenarios contain messages that wait on input from the Terminal Emulator before sending rest of the messages in the message chain. If a response from the combobox is changed when a scenario is not finished, the SIP Server Emulator must be restarted before new response configuration becomes operative.

4.2 Starting the SIP Server Emulator

To start up the SIP Server Emulator, select **Server | Start server** or click the start button. You cannot start the SIP Server Emulator twice if it is already running.

4.3 Stopping the SIP Server Emulator

To stop the SIP Server Emulator, select **Server | Stop server** or click stop button. Additionally, selecting **Server | Exit** or closing server UI window, shuts down the emulator.

When you stop the emulator, all outgoing traffic stops immediately and no new SIP messages are forwarded.

4.4 Monitoring the log

SIP Server Emulator's log can be monitored in the SIP Server Emulator GUI's log window. The new log entries are read from a log file in the Series 60 SIP Plug-in home directory `..\Series_60_SIP_Plug-in_1_0\log`.

By selecting **Enable logging** checkbox, the SIP Server Emulator starts to write a log of incoming and outgoing SIP messages and views the log in the SIP Server Emulator GUI's log window. The selection can be changed while the SIP Server Emulator is running. The log file is formed into SIP Server Emulator's home directory. The log file size is not limited. The log file should be removed every once in a while. A new log file is created automatically when starting up SIP Server Emulator if the previous log file was deleted.

If logging is disabled from the SIP Server Emulator's settings, the log is not formed and it cannot be viewed. The log file is in text format and it can also be viewed with text editors.

The log window can be closed by selecting **View | Log window** and shown again by selecting **View | Log window** or selecting and unselecting **Show log window** checkbox.

5 Using SIP Extension

SIP Extension for Series 60 SDK is installed with SIP Server Emulator. Installation program installs the SIP Extension into the Terminal Emulator's home directory. After installation it is possible to develop and test SIP applications in Terminal Emulator.

SIP Extension contains also SIP sis file, which can be found from directory [SDK_PATH]\SIP Installation.

SIP Documentation is included and is installed with SIP Extension. A link to the documentation can be found in Start menu.

6 Application installation

6.1 Target (thumb) installation

When an application using SIP is installed to terminal it needs to announce its capabilities to SIP stack. This is done from the application's pkg-file using SIPClientInstaller and SIPClientUnInstaller. To add the application information to SIP stack the following lines have to be added to application's pkg-file in the following order and according to the following instructions:

```
; This example is for thumb/urel
; 'AppName' is the application's own directory
; 'OwnPath' is the path from which the XML-file and exes can be found in
the particular environment
; XML file has to be created by each application developer
; according to the DTD provided within \Columbus\SipAppCapabilities.dtd
; An example XML can be found from:
;
\columbus\2pgaming\terminal\Battleship\Project\SipAppCapabilities.xml.
; The XML-file's name MUST be exactly SipAppCapabilities.xml.

; Application capabilities in an XML-file
"\OwnPath\SipAppCapabilities.xml"-
"!:\system\Apps\AppName\SipAppCapabilities.xml"

; Installer
"\OwnPath\SipClientInstaller.exe"-
"!:\system\Apps\AppName\SipClientInstaller.exe", FR, RI, RW

; UnInstaller
"\OwnPath\SipClientUnInstaller.exe"-
"!:\system\Apps\AppName\SipClientUnInstaller.exe", FR, RR, RW
```

And for Warship application, following excerpt of Warship pkg-file shows corresponding lines used for application installation:

```
; UNINSTALLER
"\Epoc32\release\thumb\urel\SipClientUnInstaller.exe"-
"!:\system\Apps\Warship\SipClientUnInstaller.exe", FR, RR, RW

; SIP:
"..\.Battleship\Project\SipAppCapabilities.xml"-
"!:\system\Apps\Warship\SipAppCapabilities.xml"

; INSTALLER
"\Epoc32\release\thumb\urel\SipClientInstaller.exe"-
"!:\system\Apps\Warship\SipClientInstaller.exe", FR, RI, RW
```

Note that the directives after exe-files instruct the installation system to run the installer on installation and uninstaller on uninstallation. These MUST be present. Refer SDK Help, section '» Developer Library » Tools and Utilities » Installation reference » Package file format.pkg' for more information about those attributes.

6.2 Installation under emulator (wins, winscw)

In emulator, xml file can be used as well, but it has to be copied under \epoc32\wins\c respectively, and SipClientInstaller.exe is executed in place. For example, the build script for Warship application does installation after compilation process:

```

@echo off
setlocal

set arg1=%1
set arg2=%2
set arg3=%3

pushd 2pgaming\terminal
call bldmake bldfiles
call abld build %arg1% %arg2% %arg3%
popd

if "%arg1%"=="wins" goto install
if "%arg1%"=="winscw" goto install
goto end

:install
echo Install file SipAppCapabilities.xml to \epoc32\%arg1%\c\
copy /Y 2pgaming\terminal\ Battleship\Project\SipAppCapabilities.xml
..\epoc32\%arg1%\c\
..\epoc32\release\%arg1%\%arg2%\SIPCLIENTINSTALLER.EXE
endlocal

:end
echo on

```

Appication uninstallation is managed in clean.cmd script (correct, application own SipAppCapabilities.xml assumed to be under \epoc32\wins\c during uninstall):

```

@echo off
setlocal

set arg1=%1
set arg2=%2
set arg3=%3
goto :doit

:cleanall
setlocal

cd %1
call abld clean -keepgoing %arg1%
call abld reallyclean -keepgoing %arg1%

endlocal
goto :EOF

:doit

call :cleanall 2pgaming\terminal

if "%arg1%"=="wins" goto uninstall
if "%arg1%"=="winscw" goto uninstall
goto :EOF
:uninstall
..\epoc32\release\%arg1%\udeb\SIPCLIENTUNINSTALLER.EXE

```

LIST OF TERMS AND ABBREVIATIONS

Term or abbreviation	Meaning
Ethernet Plug-in	Ethernet Plug-in for Series 60 SDK 2.0
Example SIP Application	Example SIP Application Warship
GUI	Graphical User Interface
SDK	Software Development Kit
SIP	Session Initiation Protocol
SIP Extension	Symbian SIP Stack for Series 60 SDK
SIP Proxy	A proxy server for delivering SIP messages
SIP Registrar	A server for registering SIP devices to be made available for connections
SIP Server Emulator	Combined SIP Proxy and Registrar Server
TCP	Transmission Control Protocol
Terminal Emulator	Series 60 2.0 C++ Terminal Emulator
UDP	User Datagram Protocol

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